PAYTM BUILD FOR INDIA AI HACKATHON · DELHI TRACK 1 · MERCHANT GROWTH AI

Munim

The CFO for every merchant's AI growth agent

Paytm is giving merchants AI agents that can spend money on growth. Munim decides how much they may spend, makes every offer prove itself, and stops what does not pay.

SOUNDBOX

Manzoori chahiye: 32 grahakon ko ₹250+ par ₹15 cashback, zyada se zyada ₹480.

Needs your OK: ₹15 cashback on ₹250+ bills for 32 customers, up to ₹480.

HAAN NAHI

The agent asked. Munim checked. The owner decides.

Satyam Mishra

github.com/saksham869/munim · Live demo: saksham869.github.io/munim

PROBLEM STATEMENT

Agents now spend merchants' money. Nobody is the CFO

1.57 Cr

merchants on Paytm Soundbox and devices (Q1 FY27)

Paytm is building agents for merchant marketing and customer engagement, and Pi Agents already run campaigns, loyalty and cashback for small businesses.

Give an agent a budget and three things go wrong:

Card numbers come from Munim's run on a synthetic kirana (see the live demo). Kirana margins are already thin: a distributors' federation estimates about 2 lakh kiranas shut in a year amid quick commerce (AICPDF, 2024).

OVERSPEND

Cashback bigger than the margin

₹40 back on a typical ₹341 bill at 12% margin loses money even if every customer comes back.

DEADWEIGHT

Paying people who'd come anyway

51.3% of occasional shoppers return within 14 days on their own. Most cashback sent to them is wasted.

NO PROOF

One shop can't tell what worked

With a 6–9 person comparison group, the shop's own numbers would have killed the offer that worked and kept the one that did not.

PROPOSED SOLUTION

Munim: a spending limit and a scorecard for every growth agent

1

Owner sets rules

Once, by voice on the Soundbox: ₹500 a week, keep 6% on every bill, ask me above ₹300.

2

Agent proposes

Pi Agents or any LLM suggests an offer: which customers, how much cashback, what minimum bill.

3

Munim decides

Allow it inside the rules. Hold big spends for the owner's "haan". Refuse with the exact fix.

ALLOW GATE DENY

4

Paytm runs it

A signed, single-use passport releases the offer. 20% of customers are held back to compare.

5

Munim reports

On day 14, in Hinglish on the Soundbox. Kept if it pays, stopped if not, unused budget returned.

SOUNDBOX

"Munim, sab band karo."

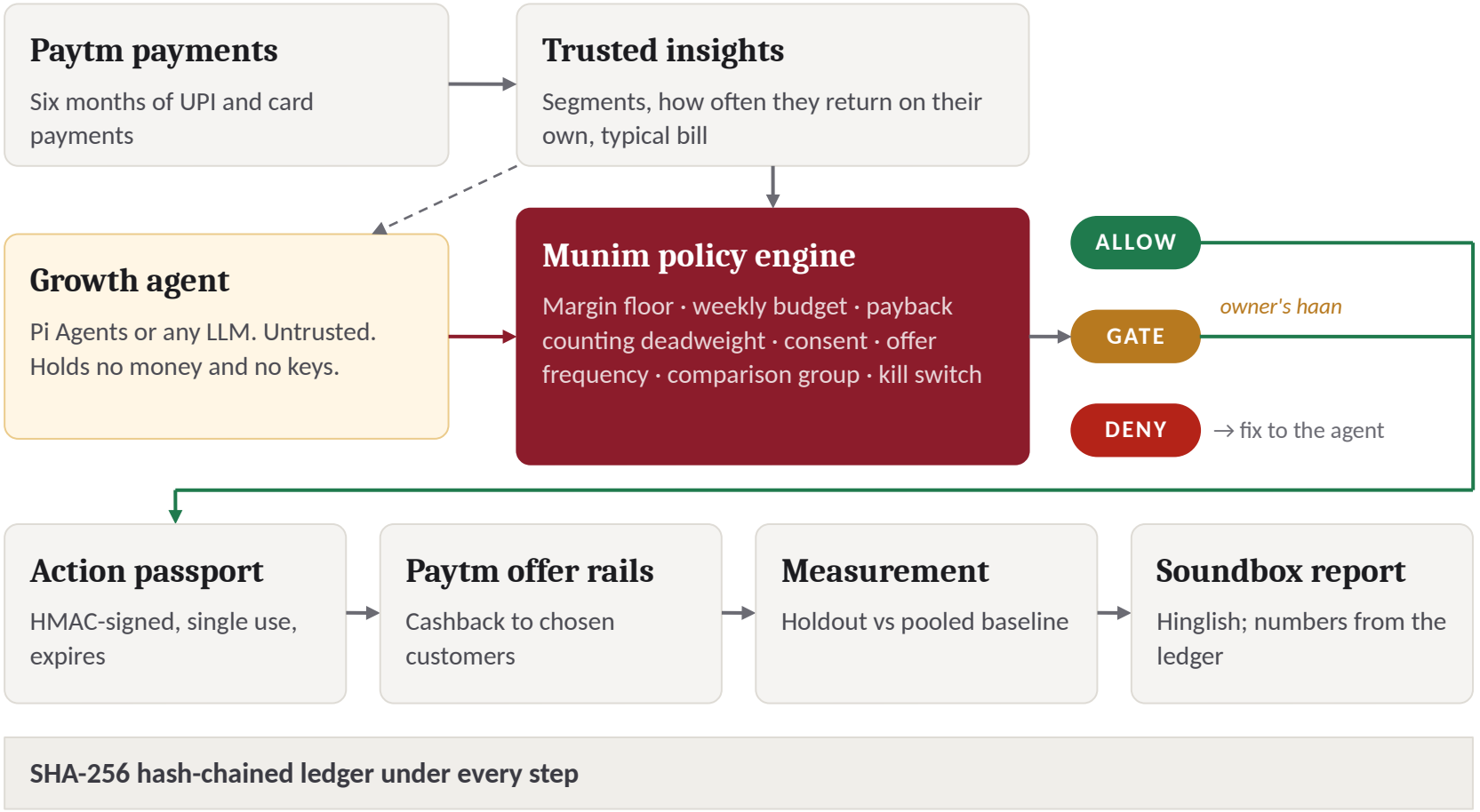
Kill switch in one sentence: from then on, every offer waits for the owner.

TAMPER-EVIDENT LEDGER

Every proposal, decision, approval, payout and result is hash-chained. Change one entry and every later hash breaks.

HOW IT WORKS

The agent proposes. Munim, with no LLM inside, decides



GUARANTEES

The agent never touches money or keys.

Insights come from Paytm data, never from the agent.

The agent can't pick who is held back: the split is salted with a secret.

Any error inside Munim means no.

Soundbox numbers come from the ledger, never from the model.

DEMO: ONE RUN, EVERY NUMBER FROM THE ENGINE

Six decisions on a synthetic kirana in Ghaziabad

Sharma General Store (synthetic): 650 customers, 13,772 payments in 6 months. 48 regulars stopped coming; they used to spend about ₹1,28,000 a month.

DENY #1

Agent: ₹40 back on any bill over ₹100, nobody held back

4 reasons, each with a fix: at most ₹6 at that bill (try ₹15 on ₹250+) · hold back 20% · ₹1,280 is over the ₹500 budget · loses money even if all return

GATE #2

Replan: ₹15 back on ₹250+, 32 customers, 9 held back

Worst case ₹480 is above the ₹300 limit. Owner says "haan"; Munim re-checks, then releases it.

DENY #3

Agent: ₹12 back for occasional shoppers

51.3% come back anyway; breaking even needs 29.0 pp more. Most it can be: ₹10.

ALLOW #4

₹10 back on ₹200+, 22 customers

Inside every rule (worst case ₹220), so it runs without asking. It is still measured.

STOPPED #5

Day 21: the ₹10 nudge

₹110 spent, 12% chance it made money. Stopped automatically; ₹110 back in the budget.

GATE #6

Kill switch: "Munim, sab band karo"

The agent's next offer (worst case ₹120) would have been allowed. Now it waits for the owner.

Ledger: 20 entries, intact. Change one verdict and the chain breaks at entry 17.

PROOF, NOT CLAIMS

One shop is too small to measure. Paytm's network is not

Same run, two ways to judge it

Offer	Truth (simulated)	Shop's own data	Munim, pooled baseline
Win back regulars	Works: +20 pp	Stop it (6% chance of profit) ×	**Keep it (93% it brings customers back) ✓**
Nudge occasional shoppers	Does nothing	Keep it (90% chance of profit) ×	**Stop it (12% chance of profit) ✓**

Why: by chance 5 of 9 held-back regulars came back anyway, and 0 of 6 occasional shoppers did. A baseline pooled from similar shops absorbs that noise.

75% right keep/stop calls (shop's own data: 68%)

24% of profitable offers wrongly stopped (45%)

1,000 simulated campaigns with known truth. Honest cost: pooling keeps slightly more losing offers (25% vs 23%).

Expected profit per campaign (₹)



With no measurement, a budget-holding agent loses money on average. Munim turns it positive.

TECHNOLOGY / TECH STACK

Small, deterministic, and ready to plug into Paytm

WHAT'S BUILT

Policy engine	Python, integer paise and basis points, no dependencies, fails closed
Measurement	Beta-binomial posteriors, pooled baselines, profit-probability verdicts
Controls	HMAC-SHA256 passports, race-safe weekly budget, kill switch
Ledger	SHA-256 hash chain, verified in one call
API	FastAPI; boots on a synthetic kirana with zero setup
LLM adapter	Any OpenAI-compatible endpoint (Pi Inference, Groq, OpenAI); output parsed, never trusted
Live demo	Same rules mirrored in JavaScript, parity-tested against the Python engine
Tests	143 automated tests (pytest)

PLUGS INTO PAYTM

Soundbox

Rules by voice, Hinglish results, "haan" to approve

Pi Agents

The proposer; Munim sits between it and the money

Payments data

Segments, return rates, typical bills

Offer rails

Merchant-funded cashback (mock adapter in demo)

WhatsApp

Customer messages through Pi's channels

USP

Guardrails stop bad prompts. Munim stops bad spending

1

Economic authority

Prompt guardrails already exist. Nothing else answers: how much may this agent spend, and did it pay?

2

Deadweight-aware

Counts cashback paid to customers who'd come anyway, and refuses offers that can't pay back.

3

Proof by default

Every offer keeps a comparison group, and the agent cannot choose who is in it.

4

Network-powered

Pooled baselines turn a 30-customer campaign into a decision. Only Paytm has that network.

5

Safe by construction

No LLM in the money path. Errors mean no. Every decision in a tamper-evident ledger.

SOUNDBOX

Agent ka offer roka, 4 kaaran. ₹100 ke bill par ₹40 cashback se har bill par 6% munafa nahi bachega. Sujhaav: ₹250+ ke bill par ₹15.

Refused, 4 reasons. ₹40 back on a ₹100 bill breaks your 6% margin. Suggested: ₹15 on bills of ₹250+.

What the owner hears when the agent overreaches.

IMPACT & BENEFITS

Growth that merchants, Paytm and customers can all trust

MERCHANTS

Growth without the guesswork

- A hard weekly cap (₹500 in the demo)
- At least 6% margin kept on every bill
- Results in plain Hindi, every two weeks
- One-sentence kill switch

PAYTM

Agents merchants keep paying for

- Measured lift to sell Pi subscriptions
- Fewer "the agent wasted my money" escalations
- Every campaign sharpens the network baseline
- A safe path to agent-to-agent payments

CUSTOMERS

Offers, not spam

- Only customers who opted in
- At most 2 offers in 30 days
- Offers sized so the shop stays healthy
- Hinglish messages, clear terms

-₹17.85 → ₹15.85

Expected profit per campaign when an agent keeps every offer, versus when Munim measures and stops the losers (1,000 simulated campaigns).

BUSINESS MODEL

Sold with Pi Agents. Paid when it works

Bundled

Munim ships inside Pi Agents' plan for small businesses. It is what makes a growth agent safe to sell to a kirana.

Outcome tier

Optional: Paytm keeps a share (say 10%) of measured net gain, only on offers Munim marks as worked.

Near-zero running cost

Decisions are arithmetic, not inference. The LLM only drafts offers and phrases results.

Beyond kiranas

The same authority layer fits Pi's banks and insurers: collection incentives, cross-sell budgets.

SCALE, ILLUSTRATED

₹18.6 Cr

a year if 1% of 1.57 Cr Soundbox merchants paid ₹99 a month

1.57 lakh × ₹99 × 12 = ₹18.65 Cr

An illustration of scale, not a forecast. It matches Paytm's stated plan to charge subscription plus inference for merchant agents.

Let the agent grow the shop. Munim keeps the money safe.

BUILT AND RUNNING TODAY

- ✓ Policy engine, measurement, passports, ledger
- ✓ 143 automated tests
- ✓ 1,000-campaign measurement evaluation
- ✓ API and a live browser demo

NEXT, FOR 31 OCTOBER

- → Hindi voice flow on the Soundbox
- → Pooled baselines from real network data
- → Real offer-rails integration
- → Approval screen for the owner

Thank you

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Demo data is synthetic. The offer API is a mock. Munim is an independent hackathon project, not a Paytm product.